Кладковой Максим 5130904/30001 Вариант 47

Формула Гаусса

$f(x) = \sin(\pi x)$; $a = x_0 = -\frac{1}{4}$; $b = x_3 = \frac{1}{4}$

$I = \int_{-\frac{1}{4}}^{\frac{1}{4}} \sin(\pi x)dx = -\frac{1}{\pi}\cos(\pi x)\Big|_{-\frac{1}{4}}^{\frac{1}{4}} = 0$

1. $I \approx \tilde{I}_1 = \frac{b-a}{2}\left[f\left(-\frac{b-a}{2}\frac{\sqrt{3}}{3} + \frac{b+a}{2}\right) + f\left(\frac{b-a}{2}\frac{\sqrt{3}}{3} + \frac{b+a}{2}\right)\right] = \frac{1}{4}\left[\sin\left(\left(-\frac{1}{4}\cdot\frac{\sqrt{3}}{3} + 0\right)\cdot\pi\right) +\right.$

$\left. + \sin\left(\left(\frac{1}{4}\frac{\sqrt{3}}{3} + 0\right)\cdot\pi\right)\right] = \frac{1}{4}\left[\sin\left(-\frac{\pi\sqrt{3}}{12}\right) + \sin\left(\frac{\pi\sqrt{3}}{12}\right)\right] = 0$

$|I - \tilde{I}_1| = 0.$

2. $I \approx \tilde{I}_2 = \tilde{I}_{21} + \tilde{I}_{22}$; $I = \int_{-\frac{1}{4}}^{0} \sin(\pi x)dx + \int_{0}^{\frac{1}{4}} \sin(\pi x)dx$

$\tilde{I}_{21} = \frac{1}{8}\left[\sin\left(\left(-\frac{1}{8}\frac{\sqrt{3}}{3} - \frac{1}{8}\right)\cdot\pi\right) + \sin\left(\left(\frac{1}{8}\frac{\sqrt{3}}{3} - \frac{1}{8}\right)\pi\right)\right] = \frac{1}{8}\left[\sin\left(-\frac{\pi\sqrt{3}}{24} - \frac{\pi}{8}\right) +\right.$

$\left.\sin\left(\frac{\pi\sqrt{3}}{24} - \frac{\pi}{8}\right)\right] \approx -0{,}09322$

$\tilde{I}_{22} = \frac{1}{8}\left[\sin\left(\left(-\frac{1}{8}\frac{\sqrt{3}}{3} + \frac{1}{8}\right)\pi\right) + \sin\left(\left(\frac{1}{8}\frac{\sqrt{3}}{3} + \frac{1}{8}\right)\pi\right)\right] \approx 0{,}09322$

$\tilde{I}_2 = 0$

$|I - \tilde{I}_2| = 0$